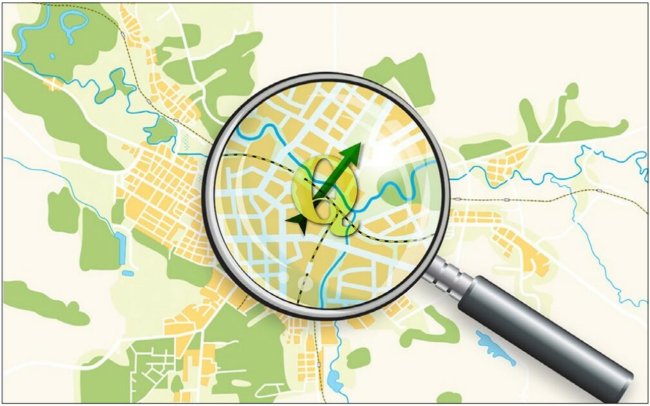
Assigning Map Projections in QGIS



QGIS is free and open source software for creating, editing, visualising, analysing and publishing geospatial information. It is OS-independent and works on Linux, Windows, Mac and BSD. An Android version is expected shortly. This tutorial on assigning map projections in QGIS is aimed at beginners.

Have you wondered how projections are assigned to particular data in GIS software? In this article, we will take our first step in exploring how map projections work in QGIS.

QuantumGIS (QGIS) is one of the world's most popular, free and open source, desktop GIS software used for making maps and in GIS analysis. The version of QGIS used in this tutorial is QGIS 1.8.0.

Map projections determine how the map being displayed is related to real places on the earth's surface. It is important that GIS software users assign a projection to the data before carrying out any analysis so that the locations in the data match the location in the real world.

Projection, or map projection, is termed as the Coordinate Reference System (CRS) or Spatial Reference System (SRS) in QGIS.

All coordinate systems used in GIS are classified into two types:

- Geographic coordinate system, and
- Projected coordinate system

A geographic coordinate system is a reference system that uses a three-dimensional spherical surface to determine locations on the earth. Any location on earth can be referenced by a point with longitude and latitude coordinates. The three dimensional surface is usually an ellipsoid. Projected coordinate systems project the ellipsoid to a 2D surface (like a sheet of paper).

QGIS allows users to define the CRS for layers without a pre-defined CRS. It also allows users to define custom coordinate reference systems and supports on-the-fly (OTF) projection of vector and raster layers. All these features allow the user to display layers with different CRS and have them overlay properly.

I have used the Karnataka administrative boundary layer in this tutorial. Whichever layer is used, the procedure will remain the same. What might change is the name of the projection assigned to the layer, which might be a different projection compared to the one used in this tutorial.